Name of the Contributer : Gauri Sandesh Dasa

DATASET : 3 1 1 _Customer_Service_Requests_Analys

PACKAGES/MODULES USED : NUMPY, PANDAS, MATPLOTLIB, SEABORN, DATETIM SCIPY

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1 .Understand the Dataset , Importing the Dataset and doing Data Preprocessing on the dataset .

In [1]:
```
# importing required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib import style
%matplotlib inline
import seaborn as sns
import datetime

from scipy.stats import chi2_contingency
from scipy.stats import chi2
```

In [2]:
```
# reading the dataset
# Note: Our Dataset is very large so setting low_memory to false
dataset = pd.read_csv('311_Service_Requests_from_2010_to_Present.csv', lo
```

In [3]:
```
# View the top 5 elements of the dataset
dataset.head(5)
```

Out[3]:

	Unique Key	Created Date	Closed Date	Agency	Agenc Nam
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New Yor City Polic Departmer
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New Yor City Polic Departmer
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New Yor City Polic Departmer
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New Yor City Polic Departmer
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New Yor City Polic Departmer

5 rows × 53 columns

In [4]:
```
# shape of the dataset
dataset.shape
```

Out[4]: (364558, 53)

In [5]:
```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 364558 entries, 0 to 364557
Data columns (total 53 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   Unique Key                      364558 non-null  int64
 1   Created Date                    364558 non-null  object
 2   Closed Date                     362177 non-null  object
 3   Agency                          364558 non-null  object
 4   Agency Name                     364558 non-null  object
 5   Complaint Type                  364558 non-null  object
 6   Descriptor                      358057 non-null  object
 7   Location Type                   364425 non-null  object
 8   Incident Zip                    361560 non-null  float64
 9   Incident Address                312859 non-null  object
 10  Street Name                     312859 non-null  object
 11  Cross Street 1                  307370 non-null  object
 12  Cross Street 2                  306753 non-null  object
 13  Intersection Street 1           51120 non-null   object
 14  Intersection Street 2           50512 non-null   object
 15  Address Type                    361306 non-null  object
 16  City                            361561 non-null  object
 17  Landmark                        375 non-null     object
 18  Facility Type                   362169 non-null  object
 19  Status                          364558 non-null  object
 20  Due Date                        364555 non-null  object
 21  Resolution Description          364558 non-null  object
 22  Resolution Action Updated Date  362156 non-null  object
 23  Community Board                 364558 non-null  object
 24  Borough                         364558 non-null  object
 25  X Coordinate (State Plane)      360528 non-null  float64
 26  Y Coordinate (State Plane)      360528 non-null  float64
 27  Park Facility Name              364558 non-null  object
 28  Park Borough                    364558 non-null  object
 29  School Name                     364558 non-null  object
 30  School Number                   364558 non-null  object
 31  School Region                   364557 non-null  object
 32  School Code                     364557 non-null  object
 33  School Phone Number             364558 non-null  object
 34  School Address                  364558 non-null  object
 35  School City                     364558 non-null  object
 36  School State                    364558 non-null  object
 37  School Zip                      364557 non-null  object
 38  School Not Found                364558 non-null  object
 39  School or Citywide Complaint    0 non-null       float64
 40  Vehicle Type                    0 non-null       float64
 41  Taxi Company Borough            0 non-null       float64
 42  Taxi Pick Up Location           0 non-null       float64
 43  Bridge Highway Name             297 non-null     object
 44  Bridge Highway Direction        297 non-null     object
 45  Road Ramp                       262 non-null     object
 46  Bridge Highway Segment          262 non-null     object
 47  Garage Lot Name                 0 non-null       float64
 48  Ferry Direction                 1 non-null       object
 49  Ferry Terminal Name             2 non-null       object
 50  Latitude                        360528 non-null  float64
 51  Longitude                       360528 non-null  float64
 52  Location                        360528 non-null  object
dtypes: float64(10), int64(1), object(42)
memory usage: 147.4+ MB
```

There are a lot of columns in our dataset but I don't need all of them. So I will drc columns that have a very large number of null values in it. Also I don't need the unnecessary columns, so I will drop them.

In [6]:
```
# viewing the columns
dataset.columns
```

Out[6]:
```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2'
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='object')
```

- Columns with most of its values as null are dropped
- Agency and Agency Name are more or less same, so I am dropping Agency Name
- Dropping other unnecessary columns like Location, Incident Adrress, Street Name, X Coord (State Plane), Y Coordinate (State Plane) as we are already considering LATITUDE and LONGITUDE
- Dropping other unnecessary columns

In [7]:
```
# dropping the irrelevant columns
drop_columns = ['Agency Name','Incident Address','Street Name','Cross Str
'Intersection Street 2','Address Type','Park Facility Name','Park Borough
'School Number','School Region','School Code','School Phone Number','Scho
'School State','School Zip','School Not Found','School or Citywide Compla
'Taxi Company Borough','Taxi Pick Up Location','Bridge Highway Name','Bri
'Road Ramp','Bridge Highway Segment','Garage Lot Name','Ferry Direction',
'X Coordinate (State Plane)','Y Coordinate (State Plane)','Due Date','Res
'Location']

dataset = dataset.drop(drop_columns, axis=1)
```

In [8]:
```
# checking the shape of our dataset
dataset.shape
```

Out[8]:
```
(364558, 14)
```

So we are upto 1 4 columns from 5 3 ...That's a lot of reduction !

- Now check for null values, data type of the columns etc.

In [9]:

```python
# checking the info of the dataset
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 364558 entries, 0 to 364557
Data columns (total 14 columns):
 #   Column                  Non-Null Count   Dtype
---  ------                  --------------   -----
 0   Unique Key              364558 non-null  int64
 1   Created Date            364558 non-null  object
 2   Closed Date             362177 non-null  object
 3   Agency                  364558 non-null  object
 4   Complaint Type          364558 non-null  object
 5   Descriptor              358057 non-null  object
 6   Location Type           364425 non-null  object
 7   Incident Zip            361560 non-null  float64
 8   City                    361561 non-null  object
 9   Status                  364558 non-null  object
 10  Resolution Description  364558 non-null  object
 11  Borough                 364558 non-null  object
 12  Latitude                360528 non-null  float64
 13  Longitude               360528 non-null  float64
dtypes: float64(3), int64(1), object(10)
memory usage: 38.9+ MB
```

In [10]:
```python
# checking the number of null values in the columns
dataset.isnull().sum()
```

Out[10]:
```
Unique Key                  0
Created Date                0
Closed Date              2381
Agency                      0
Complaint Type              0
Descriptor               6501
Location Type             133
Incident Zip             2998
City                     2997
Status                      0
Resolution Description      0
Borough                     0
Latitude                 4030
Longitude                4030
dtype: int64
```

In [11]:
```python
dataset.isnull().sum()
```

Out[11]:
```
Unique Key                   0
Created Date                 0
Closed Date               2381
Agency                       0
Complaint Type               0
Descriptor                6501
Location Type              133
Incident Zip              2998
City                      2997
Status                       0
Resolution Description       0
Borough                      0
Latitude                  4030
Longitude                 4030
dtype: int64
```

In [12]:
```
# since all the cases are of closed cases we can now drop the column Stat
dataset = dataset.drop(['Status'], axis = 1)
```

In [13]:
```
dataset.shape
```

Out[13]:
```
(364558, 13)
```

In [14]:
```
# out of the remaining columns only Descriptor, Latitude and Longitude ha
dataset = dataset[(dataset['Descriptor'].notnull()) & (dataset['Latitude'

dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 354103 entries, 0 to 364557
Data columns (total 13 columns):
 #   Column                  Non-Null Count   Dtype
---  ------                  --------------   -----
 0   Unique Key              354103 non-null  int64
 1   Created Date            354103 non-null  object
 2   Closed Date             354045 non-null  object
 3   Agency                  354103 non-null  object
 4   Complaint Type          354103 non-null  object
 5   Descriptor              354103 non-null  object
 6   Location Type           353991 non-null  object
 7   Incident Zip            354063 non-null  float64
 8   City                    354063 non-null  object
 9   Resolution Description  354103 non-null  object
 10  Borough                 354103 non-null  object
 11  Latitude                354103 non-null  float64
 12  Longitude               354103 non-null  float64
dtypes: float64(3), int64(1), object(9)
memory usage: 37.8+ MB
```

In [15]:
```
# rechecking remaining null values
dataset.isnull().sum()
```

Out[15]:
```
Unique Key                  0
Created Date                0
Closed Date                58
Agency                      0
Complaint Type              0
Descriptor                  0
Location Type             112
Incident Zip               40
City                       40
Resolution Description      0
Borough                     0
Latitude                    0
Longitude                   0
dtype: int64
```

In [16]:
```
# still there are few null values in our dataset. we will remove them
dataset = dataset[(dataset['Location Type'].notnull()) & (dataset['Incide

# rechecking for null values
dataset.isnull().sum()
```

Out[16]:
```
Unique Key                  0
Created Date                0
Closed Date                58
Agency                      0
Complaint Type              0
Descriptor                  0
Location Type               0
Incident Zip                0
City                        0
Resolution Description      0
Borough                     0
Latitude                    0
Longitude                   0
dtype: int64
```

-- So all Null values are removed from our dataset

In [17]:
```
# checking the shape our datset
dataset.shape
```

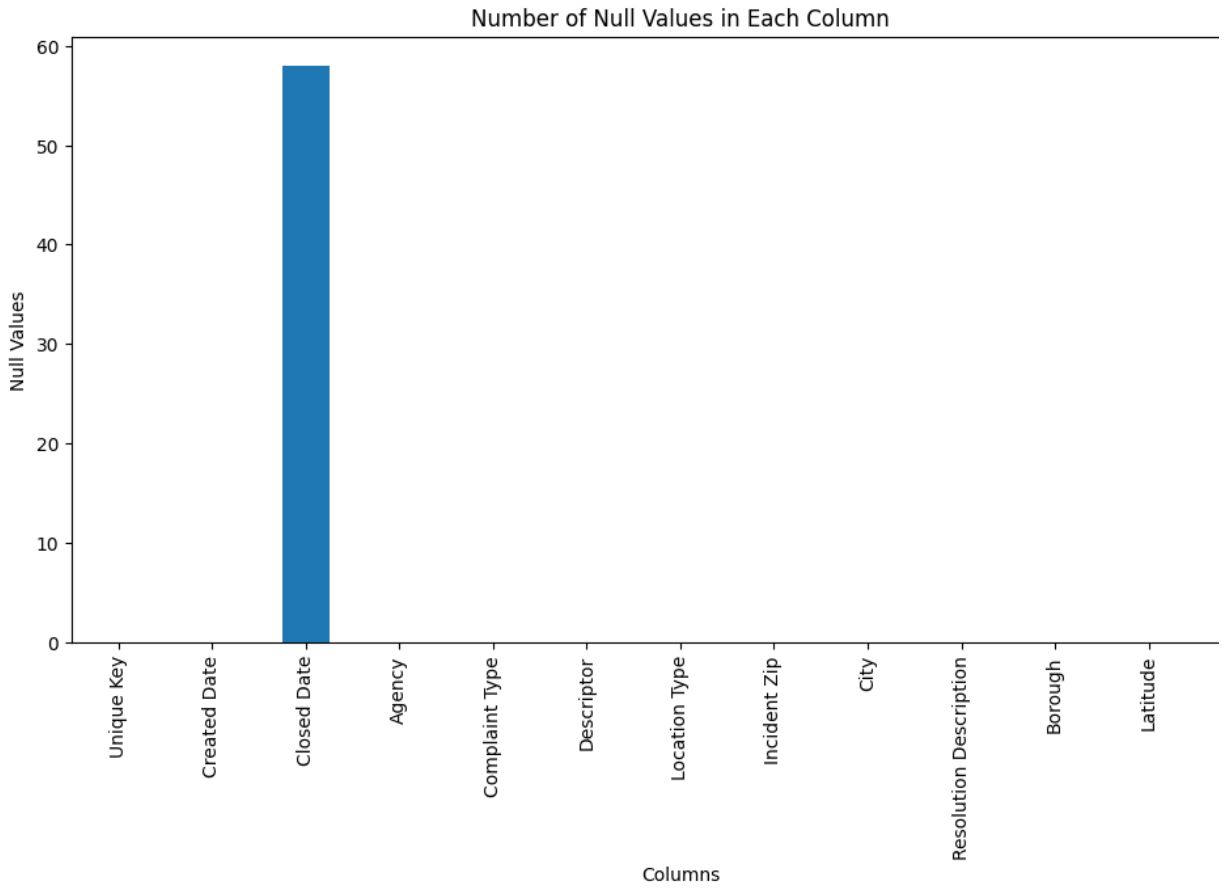
Out[17]:
```
(353949, 13)
```

2 **.basic data exploratory analysis:**

2 . 1 Draw a frequency plot for null values

In [18]:
```
null_counts = dataset.isnull().sum()

plt.figure(figsize=(12,6))
null_counts.plot(kind='bar')
plt.title("Number of Null Values in Each Column")
plt.xlabel("Columns")
plt.ylabel("Null Values")
plt.show()
```



2.2 Missing value treatment

In [19]:
```
dataset.isnull().sum()
```

Out[19]:
```
Unique Key                 0
Created Date               0
Closed Date               58
Agency                     0
Complaint Type             0
Descriptor                 0
Location Type              0
Incident Zip               0
City                       0
Resolution Description     0
Borough                    0
Latitude                   0
Longitude                  0
dtype: int64
```

2.2.1 Remove records where Closed Date is null

In [20]:
```
dataset = dataset[dataset['Closed Date'].notnull()]
```

Verify Removal

In [21]:
```
dataset['Closed Date'].isnull().sum()
```

Out[21]:
```
0
```

2.3 Analyze the date column, and remove entries that have an incorrect timeline

In [22]:
```
dataset['Created Date'] = pd.to_datetime(dataset['Created Date'])
dataset['Closed Date'] = pd.to_datetime(dataset['Closed Date'])
```

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_6424\475688951.py:1: UserWarning:
not infer format, so each element will be parsed individually, falling back t
`dateutil`. To ensure parsing is consistent and as-expected, please specify a
format.
  dataset['Created Date'] = pd.to_datetime(dataset['Created Date'])
C:\Users\Admin\AppData\Local\Temp\ipykernel_6424\475688951.py:2: UserWarning:
not infer format, so each element will be parsed individually, falling back t
`dateutil`. To ensure parsing is consistent and as-expected, please specify a
format.
  dataset['Closed Date'] = pd.to_datetime(dataset['Closed Date'])
```

Remove Incorrect Timeline Entries

Some records may have Closed Date earlier than Created Date. Remove them.

In [23]:
```
dataset = dataset[dataset['Closed Date'] >= dataset['Created Date']]
```

2.3.1 Calculate Time Elapsed

Create a new column for request closing time.

In [24]:
```
dataset['Request_Closing_Time'] = dataset['Closed Date'] - dataset['Creat
```

2.3.2 Convert Time to Seconds

In [25]:
```
dataset['Request_Closing_Time'] = dataset['Request_Closing_Time'].dt.tota
```

2.3.3 View Descriptive Statistics

In [26]:
```
dataset['Request_Closing_Time'].describe()
```

Out[26]:
```
count    3.538910e+05
mean     1.508406e+04
std      2.100190e+04
min      6.100000e+01
25%      4.531000e+03
50%      9.605000e+03
75%      1.883500e+04
max      2.134342e+06
Name: Request_Closing_Time, dtype: float64
```

This shows:

mean minimum maximum standard deviation

2.3.4 Check Null Values in Complaint Type and City

In [27]:
```
dataset[['Complaint Type','City']].isnull().sum()
```

Out[27]:
```
Complaint Type    0
City              0
dtype: int64
```

2.3.5 Replace Missing City Values

In [28]:
```
dataset['City'].fillna('Unknown City', inplace=True)
```

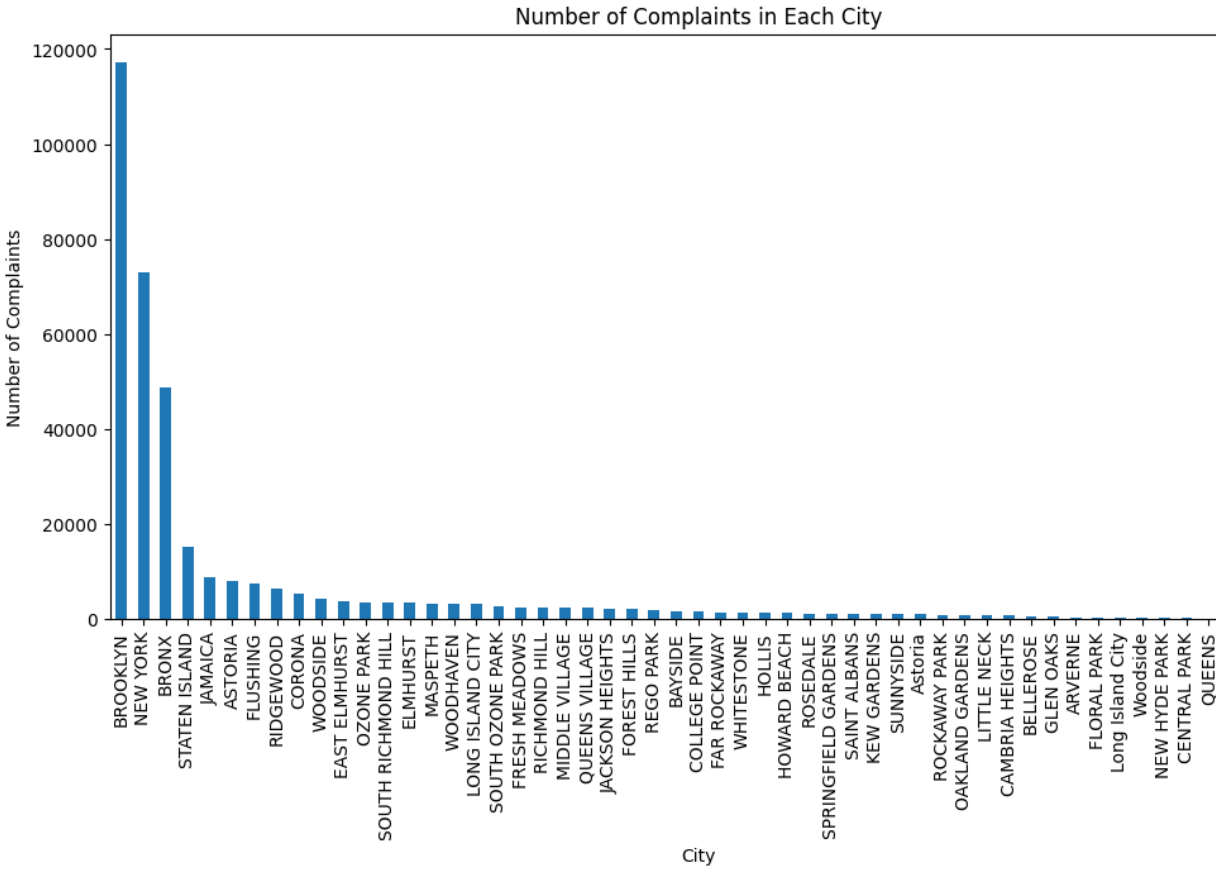
```
C:\Users\Admin\AppData\Local\Temp\ipykernel_6424\897124927.py:1: FutureWarnin
value is trying to be set on a copy of a DataFrame or Series through chained
assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work b
the intermediate object on which we are setting values always behaves as a co

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) in
to perform the operation inplace on the original object.


  dataset['City'].fillna('Unknown City', inplace=True)
```

2.3.6 Frequency Plot for Complaints in Each City

In [29]:
```
plt.figure(figsize=(12,6))
dataset['City'].value_counts().plot(kind='bar')
plt.title("Number of Complaints in Each City")
plt.xlabel("City")
plt.ylabel("Number of Complaints")
plt.show()
```

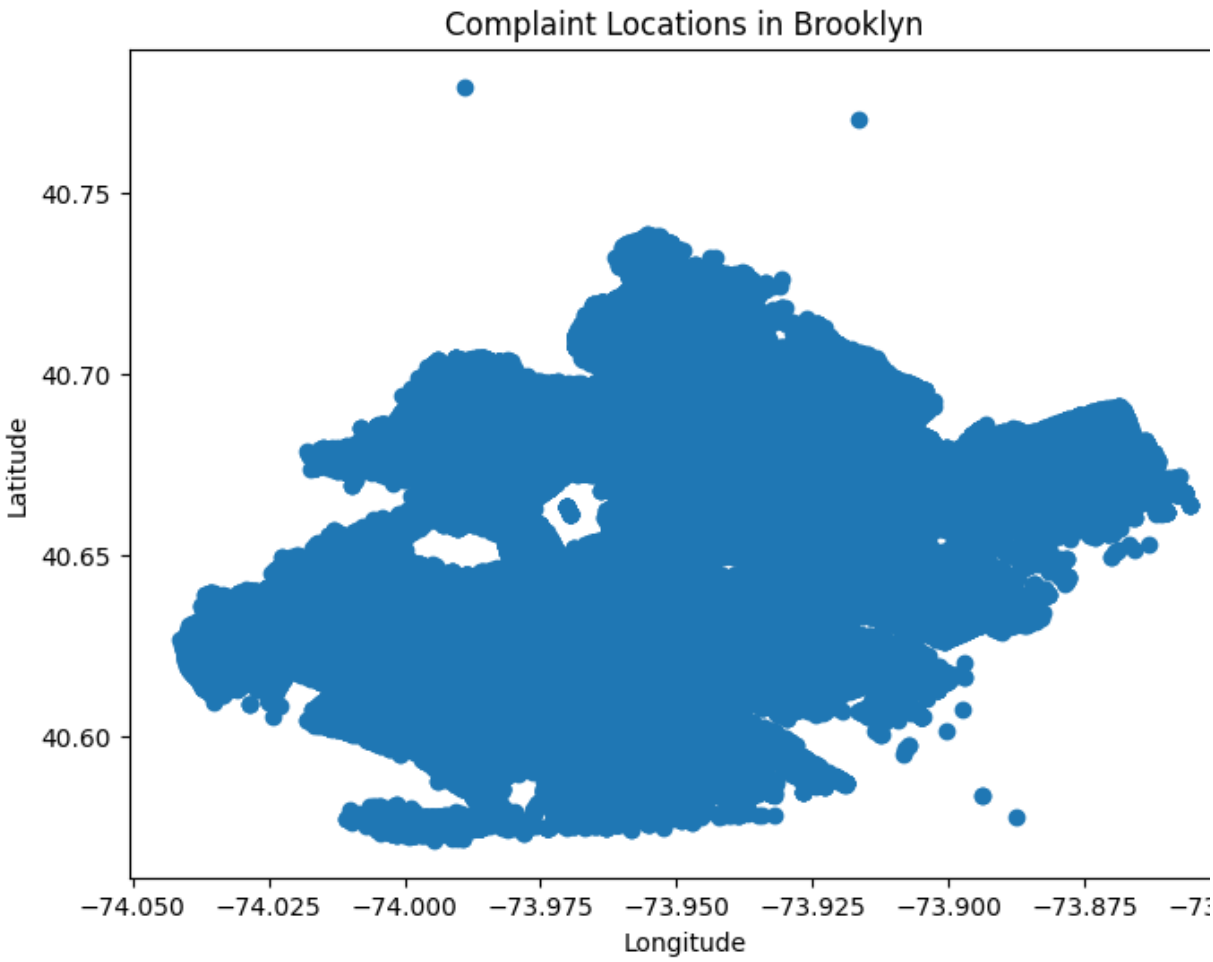


2.3.7 Scatter Plot for Brooklyn Complaints

In [30]:
```
brooklyn = dataset[dataset['City'] == 'BROOKLYN']
```

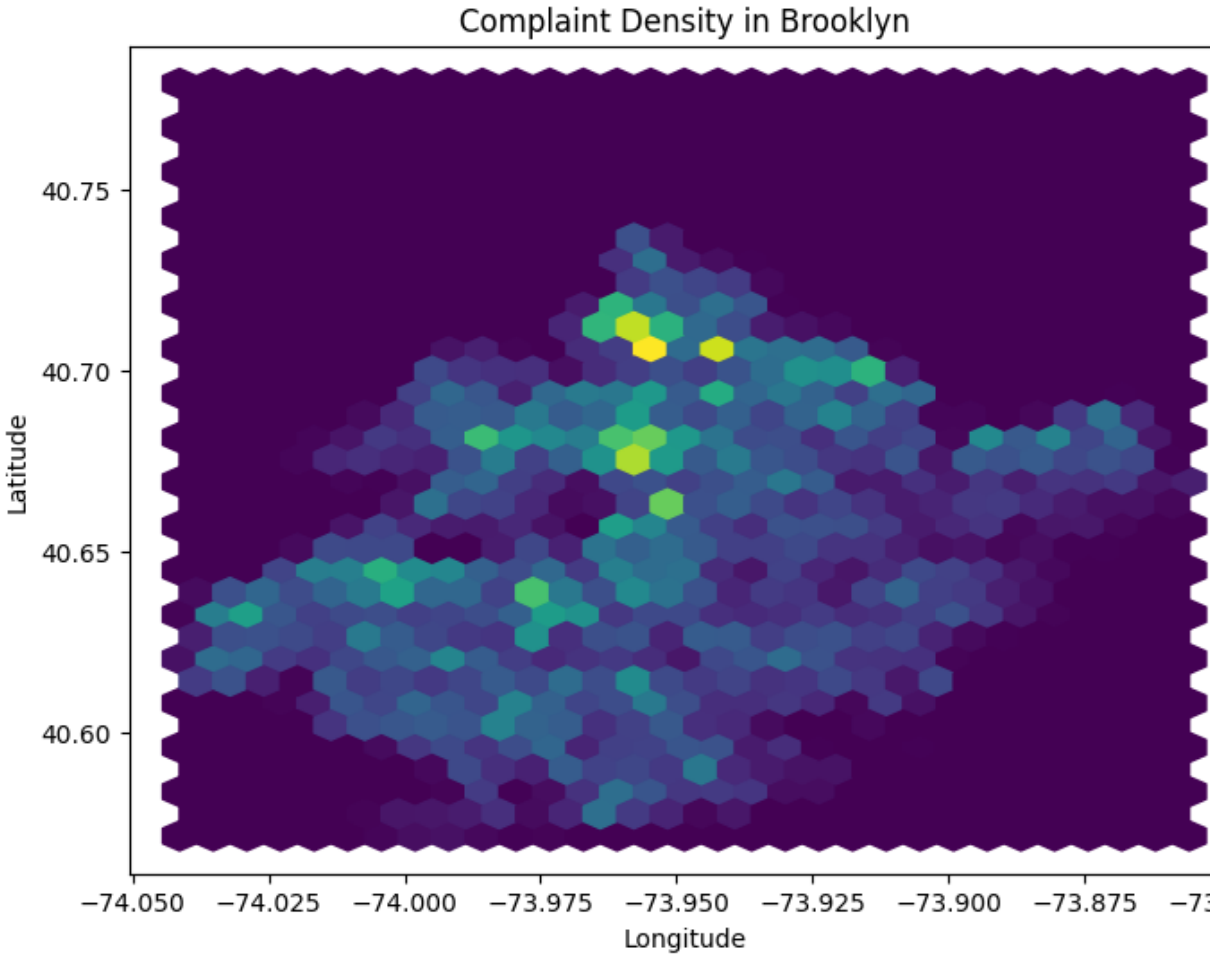
Then scatter plot:

In [31]:
```
plt.figure(figsize=(8,6))
plt.scatter(brooklyn['Longitude'], brooklyn['Latitude'])
plt.title("Complaint Locations in Brooklyn")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.show()
```



Hexbin Plot for Complaint Density

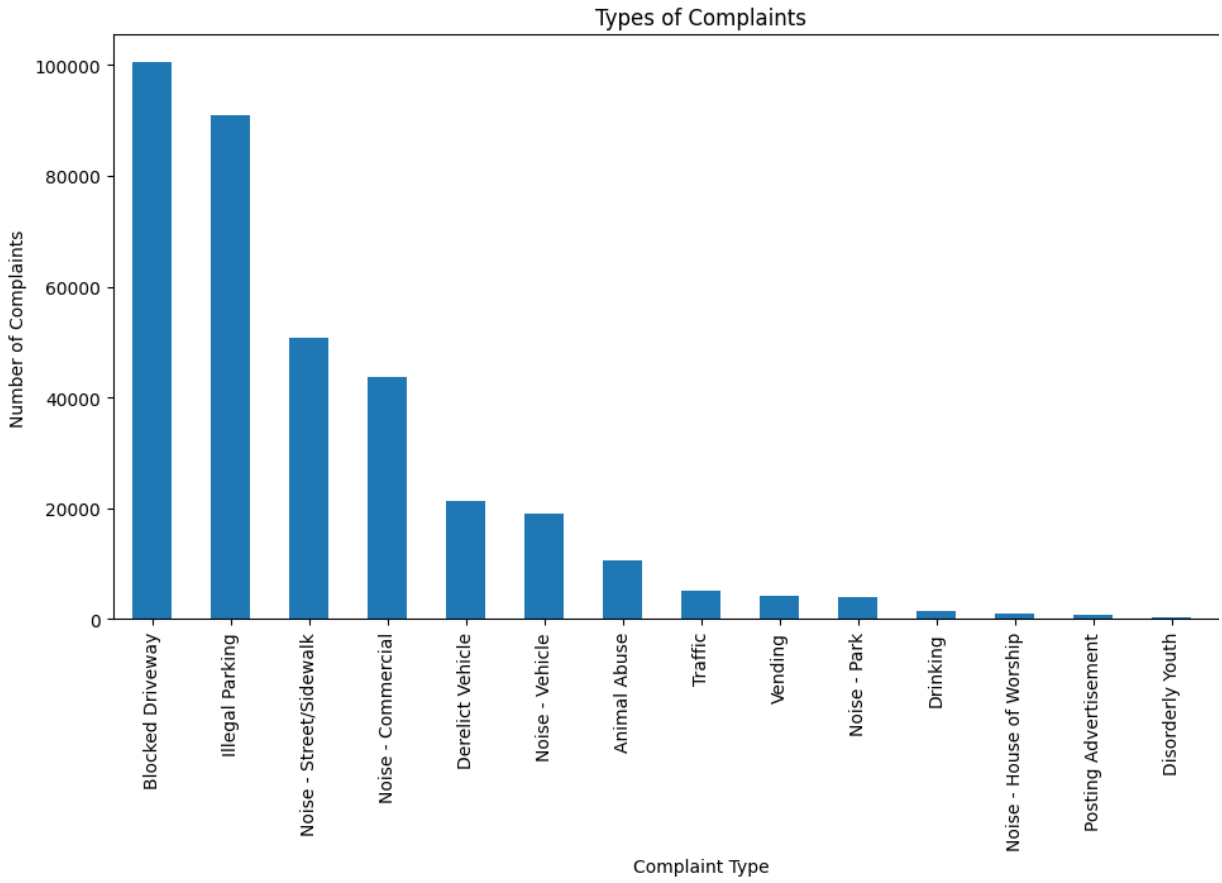
In [32]:
```
plt.figure(figsize=(8,6))
plt.hexbin(brooklyn['Longitude'], brooklyn['Latitude'], gridsize=30)
plt.title("Complaint Density in Brooklyn")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.show()
```



3.1 Plot a Bar Graph for Types of Complaints

In [33]:
```
complaint_counts = dataset['Complaint Type'].value_counts()

plt.figure(figsize=(12,6))
complaint_counts.plot(kind='bar')
plt.title("Types of Complaints")
plt.xlabel("Complaint Type")
plt.ylabel("Number of Complaints")
plt.show()
```



3.2 Check Frequency of Complaint Types for New York City

In [34]:
```
nyc = dataset[dataset['City'] == 'NEW YORK']

nyc['Complaint Type'].value_counts()
```

Out[34]:
```
Complaint Type
Noise - Street/Sidewalk     22079
Noise - Commercial          18666
Illegal Parking             14366
Noise - Vehicle              6178
Blocked Driveway             2686
Vending                      2619
Animal Abuse                 1926
Traffic                      1748
Noise - Park                 1200
Derelict Vehicle              688
Drinking                      320
Noise - House of Worship      217
Disorderly Youth               80
Posting Advertisement          49
Graffiti                       25
Name: count, dtype: int64
```

3.3 Find Top 10 Complaint Types

In [35]:
```
dataset['Complaint Type'].value_counts().head(10)
```

Out[35]:
```
Complaint Type
Blocked Driveway           100455
Illegal Parking             91057
Noise - Street/Sidewalk     50791
Noise - Commercial          43623
Derelict Vehicle            21419
Noise - Vehicle             19122
Animal Abuse                10500
Traffic                      5161
Vending                      4162
Noise - Park                 3994
Name: count, dtype: int64
```

3.4 Display Complaint Types in Each City

In [36]:
```
dataset.groupby(['City','Complaint Type']).size()
```

Out[36]:
```
City      Complaint Type
ARVERNE   Animal Abuse                46
          Blocked Driveway            50
          Derelict Vehicle            32
          Disorderly Youth             2
          Drinking                     1
                                    ...
Woodside  Blocked Driveway            27
          Derelict Vehicle             8
          Illegal Parking            124
          Noise - Commercial           2
          Noise - Street/Sidewalk      5
Length: 629, dtype: int64
```

3.5 Create DataFrame with Cities as Columns

In [37]:
```
df_new = pd.crosstab(dataset['Complaint Type'], dataset['City'])

df_new.head()
```

Out[37]:

City	ARVERNE	ASTORIA	Astoria	BAYSIDE	BELLEROSE	BREEZY POINT	BRON
Complaint Type							
Animal Abuse	46	170	0	53	15	2	1966
Blocked Driveway	50	3436	159	513	138	3	1704
Derelict Vehicle	32	426	14	231	120	3	2398
Disorderly Youth	2	5	0	2	2	0	6
Drinking	1	43	0	1	1	1	205

5 rows × 53 columns

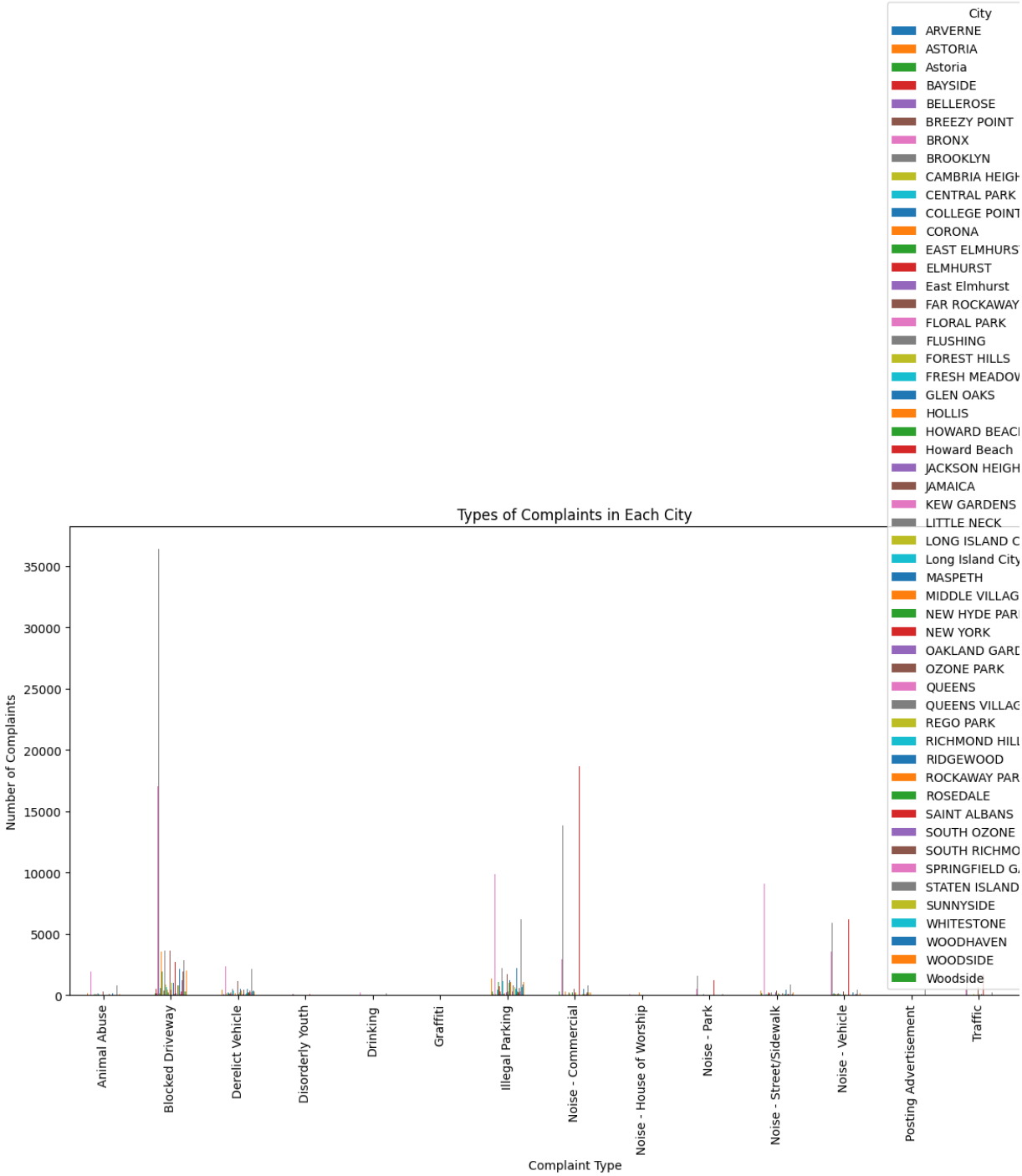
4.1 Chart Showing Complaint Types in Each City

This chart will show different complaint types with different colors.

In [38]:
```
df_new.plot(kind='bar', figsize=(15,7))

plt.title("Types of Complaints in Each City")
plt.xlabel("Complaint Type")
plt.ylabel("Number of Complaints")

plt.show()
```



This graph compares complaint types across cities.

4.2 Sort Complaint Types Based on Average Request Closing Time

Group by complaint type and calculate average response time.

In [39]:
```
avg_time = dataset.groupby('Complaint Type')['Request_Closing_Time'].mean
```

Sort the values:

In [40]:
```
avg_time_sorted = avg_time.sort_values()

avg_time_sorted.head(10)
```

Out[40]:
```
Complaint Type
Posting Advertisement        7287.171091
Noise - Commercial          11042.455562
Noise - House of Worship    11410.395656
Noise - Park                12196.250125
Noise - Street/Sidewalk     12219.275836
Traffic                     12324.517729
Disorderly Youth            12395.754777
Noise - Vehicle             12586.495921
Drinking                    13805.907791
Vending                     14355.265738
Name: Request_Closing_Time, dtype: float64
```

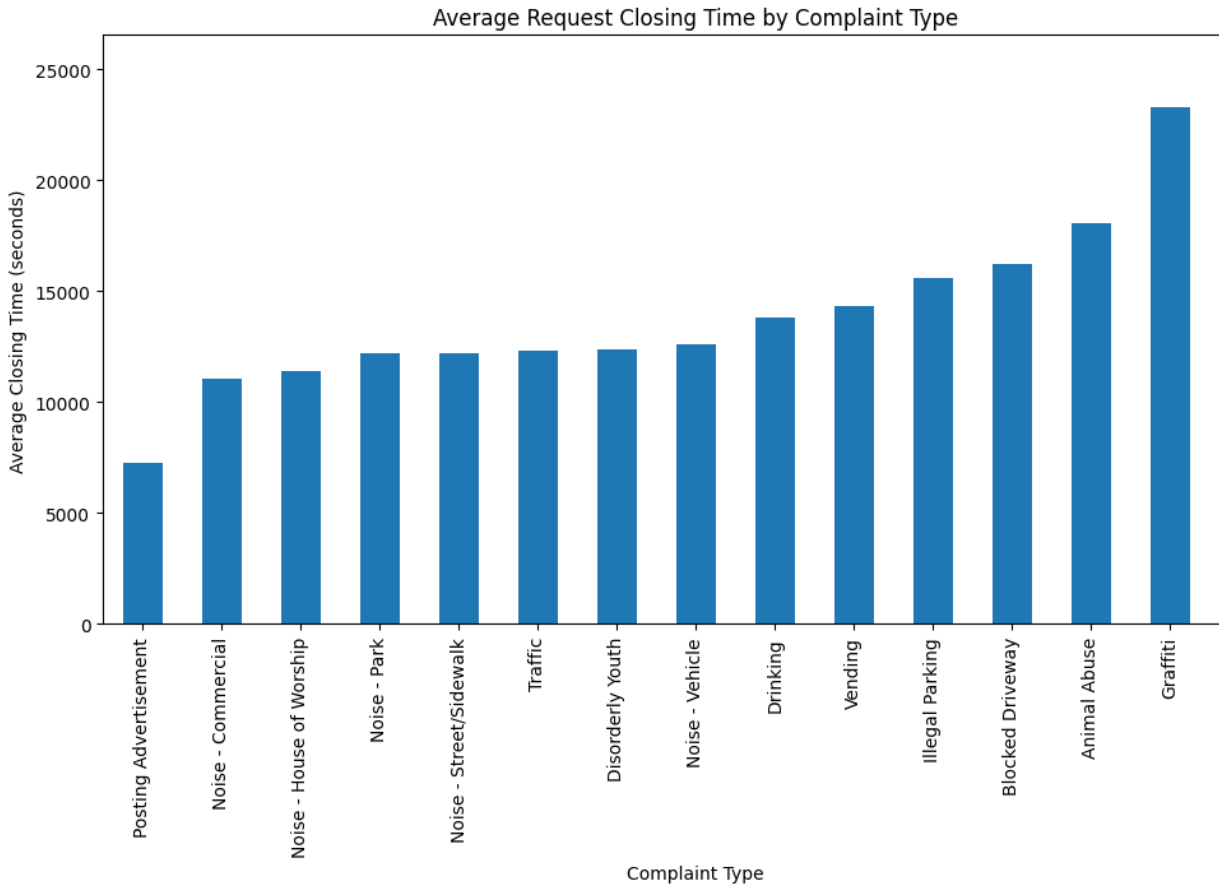
This shows complaint types with fastest response times.

Visualization of Average Response Time

In [41]:
```
avg_time_sorted.plot(kind='bar', figsize=(12,6))

plt.title("Average Request Closing Time by Complaint Type")
plt.xlabel("Complaint Type")
plt.ylabel("Average Closing Time (seconds)")

plt.show()
```



Task 5

Check whether the average response time across complaint types is similar

First calculate the average Request Closing Time.

In [42]:
```
dataset.groupby('Complaint Type')['Request_Closing_Time'].mean()
```

Out[42]:
```
Complaint Type
Animal Abuse                18052.146476
Blocked Driveway            16217.841222
Derelict Vehicle            25281.668519
Disorderly Youth            12395.754777
Drinking                    13805.907791
Graffiti                    23276.343949
Illegal Parking             15582.870960
Noise - Commercial          11042.455562
Noise - House of Worship    11410.395656
Noise - Park                12196.250125
Noise - Street/Sidewalk     12219.275836
Noise - Vehicle             12586.495921
Posting Advertisement        7287.171091
Traffic                     12324.517729
Vending                     14355.265738
Name: Request_Closing_Time, dtype: float64
```

5.1 Visualize the Average Response Time

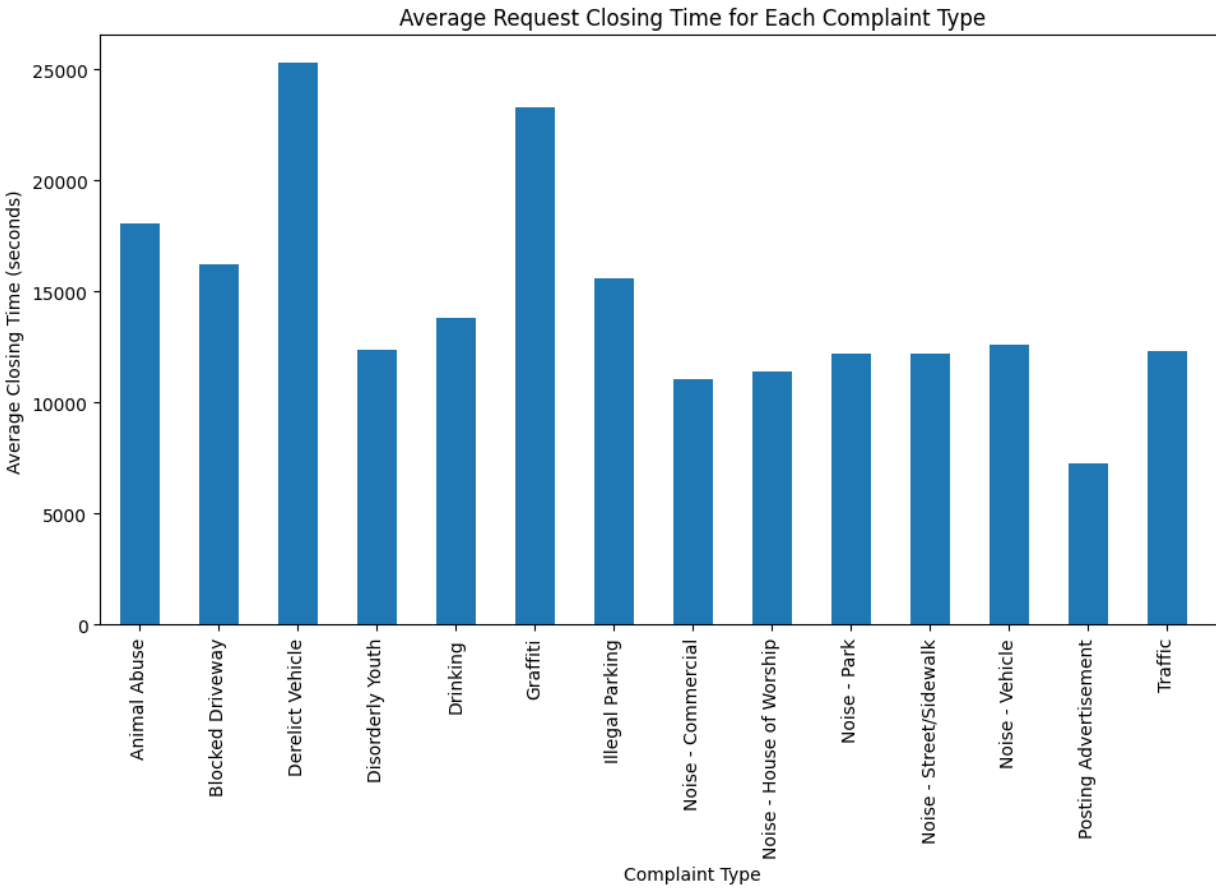
In [43]:

```
avg_response = dataset.groupby('Complaint Type')['Request_Closing_Time'].

plt.figure(figsize=(12,6))
avg_response.plot(kind='bar')

plt.title("Average Request Closing Time for Each Complaint Type")
plt.xlabel("Complaint Type")
plt.ylabel("Average Closing Time (seconds)")

plt.show()
```



This graph shows which complaints take longer to resolve.

Task 6

Identify significant variables using p-values

For this project we check whether Complaint Type affects Request Closing Time.

In [44]:
```
from scipy.stats import kruskal
```

Task 7

Perform Kruskal-Wallis H Test

Create groups of response time for each complaint type.

In [45]:
```
groups = [group["Request_Closing_Time"].values
          for name, group in dataset.groupby("Complaint Type")]
```

Now perform the test:

In [46]:
```
kruskal_test = kruskal(*groups)

print(kruskal_test)
```

```
KruskalResult(statistic=11839.696112265134, pvalue=0.0)
```

Interpretation

Check the p-value:

p-value < 0.05 → Reject H_0 → Response time differs between complaint types. p-value 0.05 → Fail to Reject H_0 → Response time is similar across complaint types.

Task 8 : Observations

You can write something like this:

Observations

```
  1.The dataset contains service request complaints from different
cities in New York.

  2. After performing data cleaning, records with missing Closed Da
were removed to ensure accurate calculation of response time.

  3. The majority of complaints are concentrated in cities such as
Brooklyn, New York, and Queens.
    Some complaint types occur more frequently than others. Noise-
related complaints are among the most common.

  4.The visualization of complaints across cities shows that differ
cities experience different patterns of complaint types.

   5. The Request Closing Time was calculated by finding the
difference between the Created Date and Closed Date, and converting
the result into seconds.
    From the analysis of average response time, some complaint type
take longer to resolve compared to others.

   6. The Kruskal-Wallis statistical test was used to check whether
the response time differs significantly across complaint types.

   7. Based on the p-value obtained from the Kruskal-Wallis test:
    If the p-value is less than 0.05, we reject the null hypothesis
and conclude that response times are significantly different for
different complaint types.

   8. If the p-value is greater than 0.05, we fail to reject the nu
```

hypothesis and conclude that response times are similar across complaint types.